

Supplemental Material for: “AI-Newton: A Concept-Driven Physical Law Discovery System without Prior Physical Knowledge”

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SM A: PHYSICAL DOMAIN SPECIFIC LANGUAGE (DSL)

AI-Newton employs a physical DSL to achieve structured and generalizable knowledge representation. This DSL builds upon expression-level syntax, formally defining both the syntax and semantics for concepts and laws. Structurally, knowledge is encoded via abstract syntax trees, where:

- Syntax rules govern knowledge definition (encoding), ensuring concepts and laws are stored in self-consistent structured forms;
- Semantic rules regulate knowledge application (decoding), enabling correct interpretation and utilization for novel knowledge discovery.

This DSL-based representation approach not only enhances AI-Newton’s expressive power for general physical knowledge but also, through the structured nature of abstract syntax trees, guarantees consistency and computability during both encoding and decoding processes.

Syntax rules for expressions

Expressions support the following binary operations:

$$P + Q, P - Q, P * Q, P / Q, P^n, \frac{dP}{dQ}, \nabla P, \quad (1)$$

where P, Q are any valid expressions.

The basic element of an expression is an atomic expression, which is a combination of a one-object concept (introduced later) with an object index, such as $v[1]$ where $v[i] := d[\text{posx}[i]]/d[t]$, or a combination of a multi-object concepts with multiple object indices, such as $\Delta[2,3]$ where $\Delta[i,j] := \text{posx}[i] - \text{posx}[j]$. Concepts that does not depend on any objects, such as time t , gravitational constant G , are also atomic expressions.

We denote the value of the expression Expr evaluated under a specific experiment ExpName as:

$$\text{eval}(\text{Expr}, \text{ExpName}), \quad (2)$$

which is computed by substituting the experimental data in the correct place.

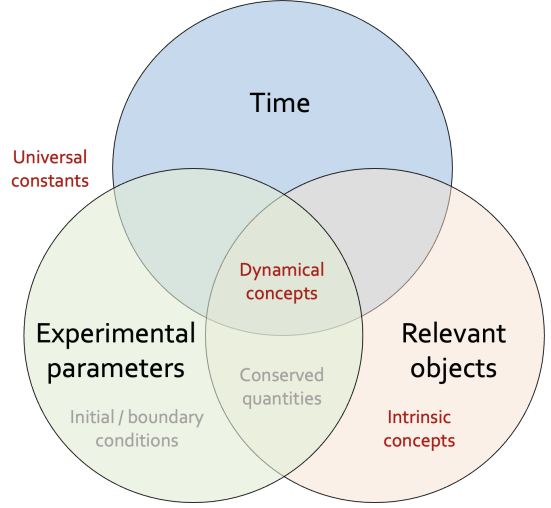


FIG. 1. **Concept types.** Physical concepts are annotated in red, while auxiliary quantities for comprehension are marked in gray.

Syntax rules for concepts

Syntax rules for concepts depend on their numerical dependencies on both relevant objects and experimental parameters, as discussed in the main text and shown in Fig. 1. Concepts are categorized into dynamical concepts, intrinsic concepts and universal constants naturally.

Dynamical Concepts: These concepts describe general properties of one or more objects and exhibit general dependence on objects and experimental parameters.

One-Object Concept: Defines a property for any single object i of a given type. The syntax is:

$$\text{Cn} := \forall i : \text{ObjectType}, \text{ExprCn}[i].$$

This rule defines $\text{Cn}[i]$ as the expression $\text{ExprCn}[i]$ for any object i of type ObjectType .

Multi-Object Concept: Generalizes the one-object concept to a collective property associated with a group of n objects, which may be of different

types. The syntax is:

$$\text{Cn} := \forall i_1 : \text{ObjectType } 1, \dots, \forall i_n : \text{ObjectType } n, \\ \text{ExprCn}[i_1, \dots, i_n].$$

This defines $\text{Cn}[i_1, \dots, i_n]$ for any set of objects i_1 to i_n of their respective types. Note that different ObjectTypes can be identical.

Summation Concept: Defines a concept as the sum of an expression over all objects of a particular type. The syntax is:

$$\text{Cn} := \text{Sum}[i : \text{ObjectType}], \text{ExprCn}[i].$$

This defines Cn as the total sum of $\text{ExprCn}[i]$ for all instances i of ObjectType .

Intrinsic Concept: Represents an inherent attribute of an object, which must be measured by evaluating an expression in a specially configured experiment. The primary syntax is:

$$\text{Cn} := \forall i : \text{ObjectType}, \\ \text{Intrinsic}[\text{ExpName}(o \rightarrow i), \text{ExprCn}].$$

This defines $\text{Cn}[i]$ for any object i by evaluating ExprCn in the context of experiment ExpName , where the generic object placeholder o is substituted with the specific object i being measured. The definition can be extended to include a standard object s for reference during measurement:

$$\text{Cn} := \forall i : \text{ObjectType}, \\ \text{Intrinsic}[\text{ExpName}(o_1 \rightarrow i, o_2 \rightarrow s), \text{ExprCn}].$$

Semantic rules for concepts

The evaluation of a concept, denoted by the function eval , depends on its type. We define the following rules:

Dynamical Concept: For a dynamical concept, its evaluation reduces to the evaluation of its underlying expression within the experimental context:

$$\text{eval}(\text{Cn}[i], \text{ExpName}) = \text{eval}(\text{ExprCn}[i], \text{ExpName}).$$

Summation Concept: For a summation concept, its evaluation is the sum of the evaluations of the expression for each object instance:

$$\text{eval}(\text{Cn}, \text{ExpName}) \\ = \sum_{i \in \text{Index}(\text{ObjectType})} \text{eval}(\text{ExprCn}[i], \text{ExpName}).$$

Here, $\text{Index}(\text{ObjectType})$ denotes the set of indices for all objects of type ObjectType present in the experiment ExpName .

Intrinsic Concept: For an intrinsic concept of an object i , its evaluation requires modifying the experimental context to specify i as the object of measurement:

$$\text{eval}(\text{Cn}[i]) \\ = \text{eval}(\text{ExprCn}, \text{ExpName}(o \rightarrow i)).$$

The notation $\text{ExpName}(o \rightarrow i)$ represents a context where the generic object placeholder o in experiment ExpName is instantiated as the specific object i .

Beyond evaluation, concepts can be transformed through specialization and generalization.

Specialization: A concept can be specialized into a set of concrete expressions for a given experiment. For example, the concept of velocity v in a two-body collision specializes to the velocities of each body:

$$\text{specialize}(v, \text{two-body collision}) = [v[1], v[2]].$$

Generalization: Conversely, an expression observed in a specific experiment can be generalized to a more abstract concept. This can result in a universally quantified concept or a summation concept. For instance:

$$\begin{aligned} &\text{generalize}(m[1] * v[1], \text{2-body collision}) \\ &= \forall i : \text{ball}, m[i] * v[i] \\ &\text{generalize}(m[1] * v[1] + m[2] * v[2], \text{2-body collision}) \\ &= \sum_{i : \text{ball}} m[i] * v[i]. \end{aligned}$$

Syntax rules for laws

We define three fundamental syntax rules for representing physical laws.

Conserved Quantity: The expression Expr is conserved in the experiment ExpName . The syntax is:

$$\text{IsConserved}(\text{Expr}, \text{ExpName})$$

Dynamical Equation (Zero Quantity): The expression Expr equals 0 in the experiment ExpName . The syntax is:

$$\text{IsZero}(\text{Expr}, \text{ExpName})$$

General Law: The concept Concept is conserved or equals zero in all experiments listed in $[\text{ValidExps}]$. The syntax is:

$$\begin{aligned} &\forall \text{ExpName} : [\text{ValidExps}], \text{IsConserved}(\text{Concept}, \text{ExpName}) \\ &\forall \text{ExpName} : [\text{ValidExps}], \text{IsZero}(\text{Concept}, \text{ExpName}) \end{aligned}$$

Semantic rules for laws

The semantic interpretation of a general law dictates how it applies to specific expressions within a valid set of experiments.

Conservation Law: A general law of type `isConserved` is interpreted as

$$\begin{aligned} \forall \text{ExpName} \in [\text{ValidExps}], \\ \forall \text{Expr} \in \text{specialize}(\text{Concept}, \text{ExpName}), \\ \text{IsConserved}(\text{Expr}, \text{ExpName}). \end{aligned}$$

Dynamical Equation (IsZero): A general law of type `isZero` is interpreted as

$$\begin{aligned} \forall \text{ExpName} \in [\text{ValidExps}], \\ \forall \text{Expr} \in \text{specialize}(\text{Concept}, \text{ExpName}), \\ \text{IsZero}(\text{Expr}, \text{ExpName}). \end{aligned}$$

SM B: RECOMMENDATION ENGINE

In AI-Newton’s autonomous discovery workflow, experiment and concept selection are governed by a recommendation engine. The engine incorporates our proposed era control strategy and utilizes a UCB-inspired value function [42–46] to maintain the exploitation-exploration balance. Specifically for concept selection, a neural network (NN) is additionally employed.

The era control strategy partitions knowledge discovery into distinct eras, each imposing an era-specific time constraint per trial (where an trial corresponds to a single analysis of one experiment). These constraints are relaxed exponentially upon era transitions. This design naturally prioritizes simpler experiments in early stages while gradually incorporating more complex ones, emulating human research practices.

At the beginning of each era, the recommendation engine resets all experiment evaluations and performs preliminary filtering to retain only feasible experiments that potentially satisfy the current era’s time limit. Subsequent selections are then made based on dynamically estimated recommendation values:

$$V(k) = \alpha R(k) + \sqrt{\frac{1}{1 + N(k)}}, \quad (3)$$

where $R(k)$ denotes the dynamic reward, $N(k)$ the historical selection count, and α the reward weighting coefficient (default $\alpha = 1.0$). This value function implements a dual-channel optimization mechanism. The exploitation channel is updated via exponential decay:

$$R(k) \leftarrow \gamma R(k) + r_t, \quad (4)$$

where when experiment k yields reward r_t , its exploitation term is updated using a discount factor γ (default $\gamma = 0.1$), which prioritizes recent outcomes to ensure adaptive response to evolving knowledge states. The exploration channel’s $\sqrt{1/(1 + N(k))}$ term creates inverse-count incentives. The engine first enforces compulsory exploration for under-explored experiments ($N(k) < 1.1$), and subsequently selects experiments with maximal $V(k)$ value during stable phases to balance the exploration-exploitation tradeoff. When $V(k) < \tilde{V}$ is satisfied for all experiments k (with preset threshold $\tilde{V}$), the engine considers all experiments to have been sufficiently analyzed, triggering AI-Newton’s transition to the subsequent era.

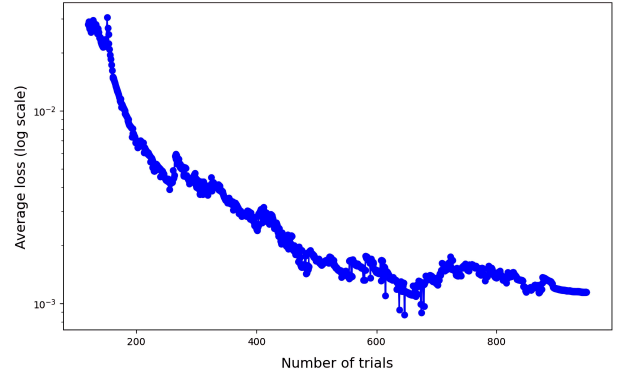


FIG. 2. **Average loss across test cases.** The average dynamic variation of loss function L_t during discovery process across 10 test cases.

Concept selection employs a value function similar to Eq. 3, with two key distinctions. First, concepts are not sampled individually but as concept groups, each containing up to three concepts. Dynamic rewards for groups are obtained by summing constituent concept rewards, while selection counts are tracked per experiment at the group level. The recommendation engine first selects top- k concept groups with highest $V(k)$ values, then randomly samples n concepts from these groups (both top- k and n are preset parameters). Second, due to concept selection’s critical and nuanced nature, reward weighting coefficient α in Eq. 3 is replaced by a fully-connected feedforward NN, denoted as α^{NN} . Its inputs include definition complexities of constituent concepts and current total number of concepts in the theory base, with single-dimensional output normalized to (0,1) via the sigmoid function. α^{NN} output intuitively represents the probability of obtaining non-zero reward when selecting concept groups with specific complexity combinations given current theory base scale. After random initialization, α^{NN} updates dynamically during the discovery process. When analyzing experiment k at trial t

with total reward r_t , contributed loss component is:

$$l_t = \frac{1}{N_{\text{grp}}} \sum_{i=1}^{N_{\text{grp}}} (\alpha_{t,i}^{NN} - \delta_{t,i})^2, \quad (5)$$

where N_{grp} denotes total concept groups involved in current trial, and $\delta_{t,i} = 0$ if $r_{t,i} = 0$ else 1 ($r_{t,i}$ represents the actual reward contribution from i -th group). Each update incorporates loss components from past T trials (default $T=256$) with relative weights ω_τ :

$$L_t = \frac{1}{T} \sum_{\tau=1}^T \omega_\tau l_\tau, \quad (6)$$

where weights are softmax-normalized with exponential decay factor η (default $\eta=0.97$):

$$\omega_\tau = \text{softmax}(\eta^{t-\tau}), \tau = t - T + 1, \dots, t. \quad (7)$$

The average dynamic variation of loss function L_t during discovery process is plotted in fig. 2.

SM C: COMPUTATIONAL ENGINE

The discovery of physical equations within our framework is driven by a two-stage computational engine designed to first generate candidate laws from data and then refine them into a minimal, non-redundant set.

The first stage employs SR to propose a wide range of mathematical expressions that fit the observed phenomena. However, this generative process can produce a large set of equations, many of which may be mathematically redundant or overly complex.

Therefore, the second stage introduces a powerful algebraic simplification mechanism. We utilize the Rosenfeld-Gröbner algorithm to analyze the differential ideals formed by the discovered laws. This allows us to systematically identify and eliminate redundancies, distilling the essential physical principles from the raw output of the symbolic search.

Differential polynomial symbolic regression.

Benefiting from the special knowledge representation, AI-Newton successfully derives critical physical laws for high-degree-of-freedom systems using only minimal SR configuration. We employ differential polynomial-level SR, which proves fully sufficient for Newtonian mechanics problems under our investigation.

The SR search space is constrained by allowed operations and concepts selected in each trial. Our SR implementation primarily involves two approaches: (1)

direct evaluation of elementary ansatz (e.g., $\alpha x + \beta y$, $\alpha xy^2 + \beta zw^2$, dx/dy , etc.), which is more crucial; and (2) principal component analysis (PCA) [47]-driven search of differential polynomial equations and conserved quantities, as analyzed in prior studies [48–50]. For general law extension, only simple term addition (Approach 1) is required, whereas direct search of specific laws utilizes both approaches.

Simplification via Rosenfeld Gröbner algorithm

Mathematical Foundation. Let $\mathcal{S} := \{a_1, a_2, \dots, a_n\}$ be the set of differential equations that are equal to 0 in specific experiment. We define a differential equation $a' = 0$ to be redundant if and only if it can be located in the root differential ideal formed by the set $\{a \in \mathcal{S} | \text{complexity}(a) < \text{complexity}(a')\}$. In the program implementation, we use the Rosenfeld Gröbner method in the differential algebra package [53] of MAPLE [54] to construct the Gröbner Basis representation of the differential equation group, and then determine whether a differential equation can be reduced by this Gröbner Basis to determine whether a differential equation is redundant.

Minimal Representation of Specific Laws. For laws discovered within a single experiment, our goal is to find the most fundamental set of governing equations, void a large number of meaningless conclusions accumulating in the theory base. We use the Rosenfeld-Gröbner algorithm to iteratively build a basis of non-redundant equations. Any proposed equation that can be derived from this basis is discarded. A newly discovered law is first checked for novelty; if it cannot be derived from the current basis of laws, it is provisionally added. After a new equation is accepted as a new member of the basis, the algorithm then re-assesses the entire, now-expanded set of laws. Starting from the most complex, if any existing equation has become redundant in light of the new addition. This self-correcting mechanism ensures the knowledge basis always represents the simplest, most fundamental set of relationships.

Reduction of General Laws. This principle of dynamic refinement is even more powerful when applied to general laws. The process is not merely additive; it is a mechanism for conceptual simplification. The introduction of a new, powerful law (e.g., one with a broad scope) triggers a re-evaluation of all existing laws. The system then systematically checks, from most to least complex, if any of these older, narrower-scoped, or more intricate laws can now be derived from the new, more fundamental principle combined with others. If so, they are pruned as redundant. This ensures that a simpler, more universal law actively replaces collections of more complex, specific ones, constantly elevating the explanatory power and elegance of the discovered physics.